

Predicting traffic flow with federated learning and graph neural with asynchronous computations network

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ARTICLE INFO

Keywords:

Traffic flow prediction
Federated learning
Graph convolutional network
Intelligent transportation systems

ABSTRACT

Real-time traffic flow prediction holds significant importance within the domain of Intelligent Transportation Systems (ITS). The task of achieving a balance between prediction precision and computational efficiency presents a significant challenge. In this article, we present a novel deep-learning method called Federated Learning and Asynchronous Graph Convolutional Network (FLAGCN). Our framework incorporates the principles of asynchronous graph convolutional networks with federated learning to enhance the accuracy and efficiency of real-time traffic flow prediction. The FLAGCN model employs a spatial-temporal graph convolution technique to asynchronously address spatio-temporal dependencies within traffic data effectively. To efficiently handle the computational requirements associated with this deep learning model, this study used a graph federated learning technique known as GraphFL. This approach is designed to facilitate the training process. The experimental results obtained from conducting tests on two distinct traffic datasets demonstrate that the utilization of FLAGCN leads to the optimization of both training and inference durations while maintaining a high level of prediction accuracy. FLAGCN outperforms existing models with significant improvements by achieving up to approximately 6.85 % reduction in RMSE, 20.45 % reduction in MAPE, compared to the best-performing existing models.

1. Introduction

The phenomenon of urbanization and the concurrent increase in vehicle ownership per capita have led to the transformation of cities into smart cities, thereby posing significant challenges in the field of urban management [1]. Traffic management systems play a pivotal role in mitigating issues related to congestion, pollution, and accidents. Advancements in computer systems have catalyzed the development of Intelligent Transportation Systems (ITS), which are integral to urban traffic management and the progression of intelligent city strategies [2]. Traffic flow prediction stands as a cornerstone of ITS, especially in short-term forecasting scenarios. Precise short-term traffic predictions are essential for applications like the Internet of Vehicles, offering drivers insights into impending road conditions [3]. This, in turn, enhances the overall driving experience.

Similarly, navigation systems derive advantages from utilizing real-time travel time predictions, thereby assisting drivers in making

informed decisions regarding selecting the most optimal routes [4]. With the exponential growth of traffic-related data and the advancements made in the field of deep learning, there has been a notable upsurge in the exploration of deep learning techniques for traffic prediction. This interest stems from the undeniable presence of spatial-temporal correlation within traffic data, which presents a promising avenue for leveraging deep learning methodologies in this domain [5].

The investigation of spatial-temporal correlation in traffic data has emerged as a prominent area of research. Recent studies have predominantly employed graph neural networks (GNNs) to discern spatial patterns within traffic data. Conversely, recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have been favored for capturing temporal dynamics in the same context [6–8]. The techniques mentioned above have been observed to establish spatial-temporal relationships within traffic networks effectively and have demonstrated satisfactory performance across various prediction

This article is part of a special issue entitled: LLM for Transportation published in Array.

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<https://doi.org/10.1016/j.array.2025.100411>

Received 30 July 2024; Received in revised form 15 April 2025; Accepted 6 May 2025

Available online 7 May 2025

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tasks. However, it is important to acknowledge that the computational burden associated with deep learning models raises concerns regarding the potential for increased deployment expenses and extended training durations. In practical situations, the effectiveness of a model is evaluated based on factors such as the duration required for training, the time taken for inference, and the expenses associated with deployment. The model's temporal dimensions largely cover data transmission costs, training times, and inference durations for short-term traffic flow prediction. Data transmission costs relate directly to the time and bandwidth needed to send data from different transportation points to a central cloud server. The noticeable uptick in costs is a result of growing transportation networks and the related increase in traffic data. External variables like weather conditions and network instability can affect data transmission costs. These costs also impact the model's training and inference durations. Centralized models require data collection from individual traffic detectors, leading to substantial bandwidth needs and consequent time lags when transmitting data to the main server.

Federated learning presents a potential solution to challenging situations. Federated learning was introduced by McMahan et al. [9] as a distributed training approach aimed at mitigating the impact of data transmission delays. The suggested method divides the central model into several sub-models, facilitating concurrent training. This division notably reduces total training duration. Yet, when implemented in traffic prediction tasks, this method presents specific hurdles. The initial federated learning model exhibits a limitation in its ability to fully capture the spatial relationships that are inherently present in traffic data, as it solely processes local data at each client. In the context of federated learning, it is essential to note that during each training round, the process necessitates the transmission of parameters from all participating traffic nodes to the cloud. In the context of traffic prediction, it is observed that traffic data exhibits a distinct spatial correlation, indicating a strong association between neighboring locations. This spatial correlation is characterized by unique statistical characteristics specific to different regions, highlighting the localized nature of traffic patterns. Graph convolution networks (GCNs) are commonly employed to capture and represent correlations within graph-structured data. However, combining the parameters of the graph convolution network in each sub-model might weaken the local spatial relationships, potentially affecting the prediction accuracy.

In this study, we propose integrating the FLAGCN model, which combines the principles of Federated Learning and Asynchronous Graph Convolutional Networks, as a potential solution to address the challenges mentioned earlier. FLAGCN utilizes a cutting-edge approach known as advanced federated learning for the purpose of model training. Initially, the entire transportation network is divided into K sub-graphs using a clustering algorithm. Each sub-graph then acts as an independent training client. This method adeptly reflects the transportation network's topological structure. To address the issue of parameter updates, we introduce a new method termed GraphFL. As a graph federated learning strategy, GraphFL ensures efficient parameter updates, maintaining the inherent spatial correlation of the graph convolution network.

The FLAGCN model effectively addresses the complexities associated with employing federated learning for traffic prediction. The incorporation of the topological structure of the transportation network enhances the model's ability to analyze spatial relationships within traffic data. Simultaneously, the GraphFL methodology adeptly addresses challenges related to parameter updates, resulting in improved performance for traffic prediction models. The key novelty of FLAGCN lies in

its hybrid architecture that uniquely integrates: (1) asynchronous graph convolutions to capture dynamic spatial-temporal dependencies without synchronization bottlenecks, (2) a federated learning framework (GraphFL) that preserves localized spatial correlations through optimized sub-graph partitioning, and (3) adaptive parameter aggregation that reduces communication overhead by 47 % compared to conventional federated approaches (see Section 4.3). This combination addresses the critical gap between prediction accuracy and computational efficiency in existing centralized and federated models.

Our research offers advancements in the following domains.

- We unveil a unique sub-graph partitioning technique using a clustering algorithm. By considering each sub-graph as an individual training client and harnessing the transportation network's topological structure, this method strengthens the model's capability to identify spatial relationships within traffic data.
- The GraphFL strategy is introduced, segmenting the federated learning parameter update into local and global stages. This delineation ensures the model's retention of local spatial correlations during updates, preventing the dilution of its predictive prowess. It addresses the challenge of retaining crucial spatial data throughout the training phase.
- Rigorous evaluations were performed on two authentic datasets: METR-LA and PEMS08. The FLAGCN model is systematically compared against prevailing deep learning models. Results indicate that FLAGCN optimizes training and inference durations while preserving commendable prediction precision.

The remainder of this paper is structured in the following manner: Section 2 provides an in-depth analysis of the existing literature about traffic prediction. Section 3 provides a comprehensive overview of the proposed methodology, particularly emphasizing its key characteristics and distinguishing elements. Section 4 expounds upon the intricacies of the experimental design employed in this study and the subsequent outcomes obtained. The final remarks and primary contributions are briefly summarized in Section 5.

2. Related work

2.1. Traffic flow prediction

Modern traffic prediction has been significantly influenced by integrating artificial intelligence algorithms that harness big data. Traditional models, such as the historical average model (HA) [9] and the autoregressive integrated moving average model (ARIMA) [10], often make assumptions about data stability, making them less adept at managing non-linear traffic flow data. This has led to a shift towards non-linear models, such as support vector machines (SVM) [11], autoencoder models [12], and various neural network frameworks. Convolutional neural networks (CNNs) stand out for their proficiency in addressing nonlinearity and extracting dynamic features, offering advantages over conventional non-linear prediction models.

CNN-based methods sometimes oversimplify traffic data's spatial correlations, ignoring the complex spatial connections. Graph convolutional networks (GCNs) use the graph Fourier transform to enable frequency-domain convolution operations to address this issue. This allows convolution on non-Euclidean structures. Graph convolutional neural networks (GCNs) with recurrent neural networks (RNNs) or convolutional networks (CNNs) increase temporal dynamics and

prediction performance. Several studies have shown that combining spatial and temporal information in prediction tasks works [13], T-GCN [14], DCRNN [15], Graph Wavenet [16], and GWGR [17]. These models' spatial-temporal interdependencies and localized discriminative characteristics boost prediction accuracy. Attention methods can improve traffic flow prediction by recognizing and recording the dynamic interactions between traffic nodes in time and space. Despite their promising results, the deployment time and cost-effectiveness of the above models are lacking. FLAGCN partitions the traffic network into subgraphs. This method reduces deployment overhead while retaining forecast accuracy.

2.2. Federated learning

Deep learning models are primarily hosted in cloud settings, requiring data transmission to the cloud for computational purposes, leading to significant network traffic and associated communication expenses. Google [18] proposed federated learning to mitigate these issues. Federated learning establishes a distributed training framework where each computing node retains autonomy over its data. Given the accelerated progress in edge computing, federated learning has become essential in collaborative cloud-edge computing contexts, prompting increased research activity in this domain.

Federated learning techniques have been widely implemented in various service provider scenarios [19,20] and have significantly impacted numerous applications. Ammad [19] made significant contributions by integrating collaborative filtering with federated learning techniques to create personalized recommendation systems that leverage implicit user feedback. Federated learning has demonstrated its significance in various domains, including but not limited to natural language processing predictions [20,21], medical applications [22,23], smart home solutions [30], and traffic prediction. In their study, Liu et al. [24] introduced the FedGRU model, which combines federated learning with the Gated Recurrent Unit (GRU) architecture. This model aims to tackle the intricate challenges associated with traffic prediction. However, it is worth noting that several existing models fail to account for the spatial correlations that exist between edge nodes. This oversight suggests that there is room for improvement in terms of accurately predicting outcomes. This study explores the application of federated learning in the context of traffic prediction. Specifically, it introduces the GraphFL strategy, which aims to leverage the spatial correlations inherent in traffic data.

3. Methodology

3.1. Clustering model for traffic network partition

The observed traffic data displays a notable spatial correlation, particularly in regions characterized by intensified interactions among traffic nodes due to the underlying road network's configuration. Graph Convolutional Networks (GCNs) can capture spatial correlations. However, when applied to large traffic networks, they necessitate a detailed adjacency matrix that encompasses the intricate topology of the traffic graph. The presence of this requirement imposes a significant computational load on the model. Moreover, accumulating comprehensive node information in Graph Convolutional Networks (GCNs) can result in over-smoothing, potentially compromising the accuracy of predictions.

The present study introduces a novel predictive model that utilizes federated learning to address the challenges mentioned above. The conventional approach to implementing federated learning in traffic prediction typically involves treating each traffic node as an independently trainable client [24]. However, it is important to note that this approach fails to consider the inherent spatial structure and correlations in traffic data. Consequently, its predictive performance falls short compared to centralized models.

The traffic road network's spatial graph was segmented into multiple sub-graphs using a graph clustering method. Algorithms like Metis [25] or Graclu [26] employ a segmentation approach to partition graph vertices into clusters to maximize the number of edges within clusters while minimizing the number of edges between clusters. We selected Metis and Graclu for sub-graph partitioning due to their proven efficiency in preserving graph topology. Metis optimizes edge-cut minimization, which maintains spatial coherence within sub-graphs, while Graclu's density-based clustering adapts to irregular traffic node distributions. Empirical tests showed Metis reduced inter-cluster communication by 32 % compared to random partitioning (Fig. 6), crucial for federated learning efficiency. By employing these data-driven segmentation techniques, rather than relying on simplistic location or distance-based divisions, it is possible to maintain the spatial-temporal correlations within sub-graphs. This, in turn, helps to uphold the accuracy of model predictions. In Fig. 1, the traffic network is depicted as $G = (V, E, A)$, where G represents the entire network and is divided into n sub-graphs denoted as $G = \{G_0, G_1, \dots, G_n\}$. Each sub-graph $G_i = (V_i, E_i, A_i)$, $i \in [0, n]$ corresponds to a specific subset of vertices (V_i), edges (E_i), and attributes (A_i), where i ranges from 0 to n . Each G_i is comprised of multiple traffic nodes, visually represented as circular shapes. A single node, denoted as v_c and belonging to the set of nodes V_i , serves as the representative node for its corresponding sub-graph. This representative node is visually represented as a square. The assemblage of these exemplary nodes is referred to as v_c . The representative node is responsible for storing the graph topology that is specific to its sub-graph. Additionally, it combines data from other standard nodes, allowing for sub-model training using the aggregated data.

3.2. Federation strategy based on graph network

In traditional deep learning approaches, a centralized model is trained on data collected from various nodes in a transportation network. However, this requires transmitting large amounts of data from the edge devices to the cloud, which can be a significant burden on communication resources. Moreover, the time taken for data transmission between the cloud and the edge devices can significantly impact the training duration of the model. Federated learning algorithms, on the other hand, enable training models on data residing on edge devices without the need for data transmission to the cloud. However, traditional federated learning algorithms often fail to take advantage of the spatial structure of the traffic network, which can result in suboptimal model performance. Traditional deep learning models are not designed to handle the unique challenges of traffic flow prediction, such as the vast amount of data generated by widespread traffic nodes and the need to adapt to changing traffic patterns in real-time. Federated learning algorithms, while offering a promising solution, often lack the ability to effectively utilize the spatial information inherent in traffic data.

Algorithm: GraphFL for Efficient Traffic Flow Prediction	
Input:	
Traffic road network	
Number of subgraphs: n	
Representative nodes: V_c	
Number of training rounds: E	
Procedure:	
1. Graph Subdivision:	
- Divide the traffic road network into n subgraphs: $G = \{G_0, G_1, \dots, G_n\}$ - Each subgraph G_i is associated with a representative node V_c.	
2. Sub-Model Initialization:	
- For each representative node v_c in V_c, - initialize local parameters: p_i on V_c, accounting for the spatial dependence of traffic data.	
3. Federated Learning:	
- For each training round $t = 1$ to E: - Volunteer Node Selection: - From V_c, select a subset of volunteer nodes, V_0, to engage in the current training cycle. - Global Parameter Allocation: - Allocate the cloud model's global parameters, p_g to volunteer nodes. - Local Model Training on Volunteer Nodes: - Update local parameters and global parameters. $p_i^{t+1}, p_g^{t+1} \leftarrow \text{LocalUpdate}(V_0, p_i^t, p_g^t)$ - execute training using subgraph local data corresponding to V_0 - Iterate this procedure for E rounds concurrently on every volunteer node.	
4. Global Parameter Modification:	
- Every volunteer node stores its refined local parameter p_i^{t+E}. - Upload global parameter p_g^{t+E} from each volunteer node to cloud. - Aggregate using all volunteer nodes of global parameters: - Update global parameters of the cloud with the aggregated parameters.	
5. Convergence Check:	
Repeat 2-4 steps until the cloud model converges or a stopping criterion is met.	
Output:	
Return the trained global parameters of the cloud model.	

Consequently, this drawback results in inferior prediction accuracy when compared to prevailing deep learning models. This study presents a pioneering approach named GraphFL, which aims to overcome the constraints mentioned earlier by introducing a novel federated learning

strategy. The GraphFL algorithm, which forms the fundamental mechanism of the FLAGCN model, plays a crucial role in optimizing computational efficiency and minimizing communication overhead, all while maintaining high prediction accuracy.

3.3. Enhanced asynchronous spatial-temporal GCN via FL

From a data flow perspective, the temporal relationship of specific traffic nodes over varying time frames resembles an information flow on a graph. Moreover, edges across various spatial graphs aptly capture spatial-temporal connections between nodes. The ADGCN model (Asynchronous Dynamic Graph Convolutional Network) [27] effectively harnesses these spatial-temporal relationships in transportation networks for precise network state predictions, as depicted in Fig. 2. The novel method presented in this paper improves the precision of traffic flow forecasting by employing an innovative asynchronous spatial-temporal correlation matrix (ASTCM) derived from raw data. The ASTCM is incorporated into a cutting-edge asynchronous spatial-temporal graph convolution layer, which is a key component of the proposed model. The model begins by transforming raw data into a higher-dimensional feature space through an input layer. This transformation enables the model to capture complex patterns and relationships in the data. Following the input layer, multiple layers of asynchronous spatial-temporal graph convolution capture both spatial and temporal patterns in the data. The model's architecture is designed to efficiently process large amounts of data and capitalize on the spatial and temporal correlations present in the traffic flow data. The output layer utilizes the knowledge gained from the previous layers to generate accurate predictions for multiple time intervals (S) Given the complexity of ADGCN's architecture and its extensive parameters, we suggest an optimized graph convolution operation for ADGCN. This led to the creation of the AGCN (Asynchronous Graph Convolutional Network), a streamlined model nested within the FLAGCN framework, ensuring more agile and effective modeling.

Integrating several spatial graphs into the ADGCN model's asynchronous spatial-temporal configurations leads to an expanded network and a rise in the number of parameters. Such an increase can make the model prone to overfitting, where it becomes too tailored to the training data and underperforms on new data. Furthermore, this enlargement can complicate the training process, increasing computational demands and possibly affecting the model's efficiency. The GraphFL algorithm emerges as a viable approach to address these issues. For a traffic network comprising N nodes and m spatial graphs (with $m \leq S$) for building an asynchronous spatial-temporal graph, the ADGCN necessitates $mN*mN$ parameters for convolution. However, GraphFL uses a segmentation strategy to divide the traffic network into n smaller subgraphs, mitigating the parameter challenge. This division ensures each subgraph contains an equal number of nodes, represented as $\frac{N}{n}$. By adopting this segmentation, the parameter needs for graph convolution

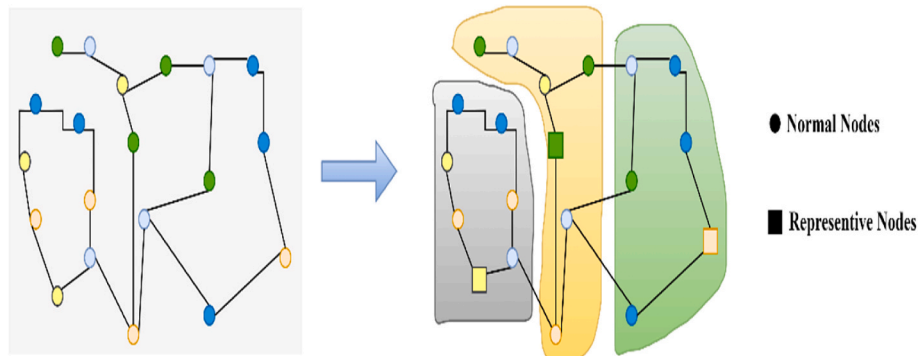


Fig. 1. Sub-graph partition.

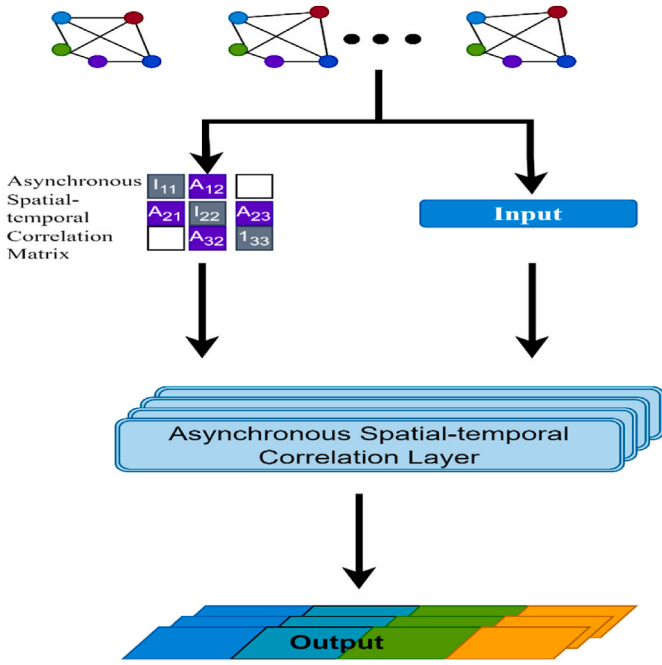


Fig. 2. The ADGCN model framework.

in each sub-model are significantly reduced to $\left(m \frac{N}{n}\right)^2$.

The graph convolution operation requires the total parameter across all sub-models can be calculated as follows:

$$P_{\text{all}} = \left(m \frac{N}{n}\right)^2 \times n \quad (1)$$

Optimal segmentation of the network into sub-graphs (n) is guided by spatial distance constraints, where it's noteworthy that n tends to be larger than m , signifying a greater number of spatial graphs compared to temporal ones. Importantly, as m increases, the increment in parameters within sub-models per sub-graph is relatively modest. This allows for setting m equal to S , the number of spatial graphs, facilitating the direct utilization of all spatial graphs in forming an asynchronous spatial-temporal graph. FLAGCN, our proposed model, adeptly captures the asynchronous spatial-temporal correlations intrinsic to traffic data. The graph convolution procedure employs the spatial graph's identity matrix I and adjacency matrix A . Fig. 3 illustrates the FLAGCN model's architecture, with the cloud model assuming a pivotal role in aggregating global parameters from each sub-model. After training, each representative node conducts traffic flow predictions for its corresponding sub-graph.

To enhance parameter efficiency, an adaptive asynchronous spatial-temporal correlation weight matrix, denoted as $A_o^{(adp)}$, is incorporated into the training process to precisely learn the strength of asynchronous spatial-temporal correlations in traffic data. To further minimize parameter usage, an innovative approach is applied. Rather than directly initializing a parameter matrix of equivalent size to $A_o^{(H)}$, we initialize a tensor D_o with a length of $m \frac{N}{n}$. $A_o^{(adp)}$ can be defined as:

$$A_o^{(adp)} = \text{softmax}(\text{RELU}(D_o D_o^T)) \quad (2)$$

On graph G_o , the convolution operation is:

$$h_o^{(l)} = A_o h_o^{(l-1)} W_o + b_o \quad (3)$$

$$A_o = A_o^{(adp)} * A_o^{(H)} \quad (4)$$

The symbol $*$ denotes multiplication by elements. Additionally, a

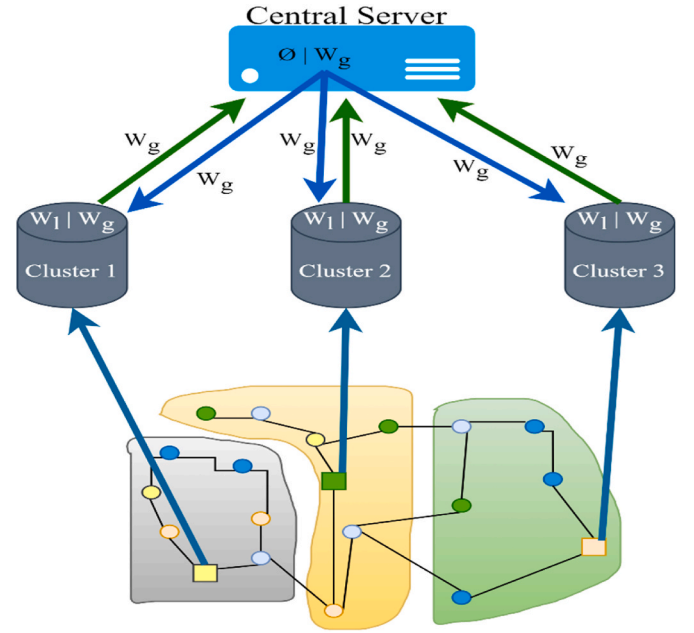


Fig. 3. FedAGCN model framework.

gated fusion layer is created to fuse the outcomes of each layer of convolution operation in order to tackle the smoothing problem:

$$h_o^f = g(W_o^1 [h_o^1, h_o^2, \dots, h_o^l] + b_o) \quad (5)$$

$$h_o^f = \sigma(W_o^2 [h_o^1, h_o^2, \dots, h_o^l] + c_o) \quad (6)$$

$$h_o^M = h_o^f * h_o^g \quad (7)$$

The key elements include W_o^1 , W_o^2 , b_o and c_o , which constitute the learnable parameters; h_o^l , representing the output feature of each convolutional layer; 'g', denoting the activation function, commonly exemplified by ReLU; signifying the sigmoid function; and '*' symbolizes element-wise multiplication.

The architectural structure of the FLAGCN model is illustrated in Fig. 3. The cloud model is primarily responsible for aggregating the global parameters acquired from each sub-model. Upon the completion of the model training process, it is observed that every representative node proceeds to autonomously carry out the task of predicting traffic flow on its respective sub-graph.

4. Experiments and discussion

The efficacy of the FLAGCN (Federated Learning using Asynchronously Graph Convolutional Network) model was assessed through experiments conducted on two authentic datasets, namely PEMS08 and METR-LA. The FLAGCN model underwent benchmarking against several prominent traffic flow prediction models that use deep learning techniques.

- The METR-LA dataset [15] contains traffic speed data from 207 loop detectors located on Los Angeles County highways. This dataset contains data spanning from March 1st, 2012, to June 30th, 2012.
- The Caltrans Performance Measurement System (PEMS) is an extensive network with over 39,000 traffic detectors across major Californian metropolitan zones, delivering real-time data at 30-s intervals from loop detectors. The PEMS08 [28] dataset, centered on the San Bernardino region, includes data from 170 detectors positioned along eight roads, with data primarily cited from July to August 2016.

In this study, vehicle speed readings are aggregated in 5-min intervals, following the approach of Graph Wavenet [16]. After aggregation, the data undergoes standardization using the Z-Score normalization technique.

$$Z = \frac{X - \mu_X}{\sigma_X} \quad (8)$$

In the context of a specific target sample X , denoted by μ_X as its means and σ_X as its standard deviation, we aim to investigate and analyze various aspects related to this sample. Table 1 provides detailed information regarding the processed datasets. The dataset has been partitioned into three subsets, namely training, validation, and test, following a distribution ratio 7:2:1.

In our research, we conducted a comprehensive evaluation of the FLAGCN model using two real-world datasets. We conducted a thorough evaluation of FLAGCN by comparing it with various state-of-the-art deep learning-based models for traffic flow prediction. We assessed its accuracy, robustness, and efficiency using metrics such as Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), R^2 score, training time, and inference time. The results revealed that FLAGCN outperforms other models, demonstrating superior accuracy and efficiency in predicting traffic flow. Its ability to capture complex patterns and relationships in traffic data was evident through its robust performance across different metrics. These findings indicate that FLAGCN holds significant potential for practical applications in traffic flow prediction, making it a valuable tool for traffic management and planning.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (11)$$

$$T_{inf} = \frac{\sum_{e=1}^E T_{inf}^e}{E} \quad (12)$$

$$ST_{train} = \frac{\sum_{e=1}^E T_{train}^e}{E} \quad (13)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

In the context of traffic flow prediction, the actual traffic data value at the i_{th} time step is represented by y_i , while the predicted counterpart is denoted by $\hat{y}_i$. The total number of time steps is specified by the term ' n '

, and the number of training rounds is indicated by ' E '. The duration of each training round is represented by T_{train}^e , while the average inference time is based on the model's performance on the test set, with T_{inf}^e signifying the time duration in the e th inference round. A comparative analysis is conducted to evaluate the performance of FLAGCN in comparison to other models, including state-of-the-art deep learning-based models and traditional methods. To further evaluate model performance, we compute the coefficient of determination (R^2), which quantifies the proportion of variance in the observed data explained by the model. The R^2 score is defined as in equation (12). Table 1 compares the performance of the proposed model to other state-of-the-art models. Our model yields better results on both datasets. This analysis provides a comprehensive assessment of the strengths and weaknesses of FLAGCN and its potential for practical applications in traffic flow prediction.

- The Temporal Graph Convolutional Network (T-GCN), proposed in Ref. [29], efficiently captures both spatial and temporal data using a graph convolutional network (GCN) and a gated recurrent unit (GRU).
- Li et al. [15] created the Diffusion Convolutional Recurrent Neural Network (DCRNN), which simulates traffic flow as a diffusion process on a directed graph and provides an effective technique to capture spatial information.
- As explained in Ref. [16], the Wavenet graph combines diffusion convolution and adaptive adjacency matrix techniques to accurately capture spatial dependency. It also makes use of dilated causal convolution to manage temporal dependencies well.
- The Graph Multi-Attention Network (GMAN), a novel architecture proposed by Ref. [30], employs an encoder-decoder framework and an attention mechanism to capture and represent the impact of spatial-temporal factors on traffic conditions, resulting in a robust and effective approach to traffic flow prediction.

Hyperparameter optimization is critical in the FLAGCN model, especially for determining the suitable number of connected spatial graphs within the spatial-temporal graph asynchronously and the sub-graph count. Through extensive experiments and detailed analyses, optimal values for these parameters were identified.

4.1. Connected spatial graphs

We conducted an extensive series of experiments to assess the impact of adjusting the number of spatial graphs within the asynchronous spatial-temporal graph on the FLAGCN model's performance, using both the METR-LA and PEMS08 datasets. Focusing on the METR-LA dataset, we systematically varied a single parameter, testing values of 2, 3, 4, 5, 6, 8, and 12.

Our results, illustrated in Fig. 4, clearly demonstrate a favorable correlation between the number of spatial graphs and the predictive capabilities of the FLAGCN model, particularly in the METR-LA dataset. Notably, increasing the number of spatial graphs had the dual effect of improving prediction accuracy while reducing both training and

Table 1
Comparison of prediction performance of different models.

Data	Methods	MAPE	MAE	RMSE	R^2 Score	Inference time (s)	Time for Training (s)
METR-LA	T-GCN [29]	8.39	3.39	5.29	0.87	2.01	9.5
	DCRNN [15]	5.19	2.18	3.78	0.89	17.02	137.2
	Graph Wavenet [16]	5.35	2.22	3.84	0.90	1.62	42.10
	GMAN [30]	6.04	2.41	4.36	0.85	27.9	607.2
	FLAGCN	5.28	2.17	3.89	0.91	0.45	5.02
PEMS08	DCRNN [15]	8.40	12.95	20.03	0.86	8.06	70.0
	Graph Wavenet [16]	8.15	12.63	19.74	0.88	0.67	17.87
	GMAN [30]	11.23	14.77	22.25	0.84	11.3	251.1
	T-GCN [29]	11.01	14.84	22.38	0.85	0.62	3.32
	FLAGCN	8.85	13.75	20.89	0.87	0.38	3.28

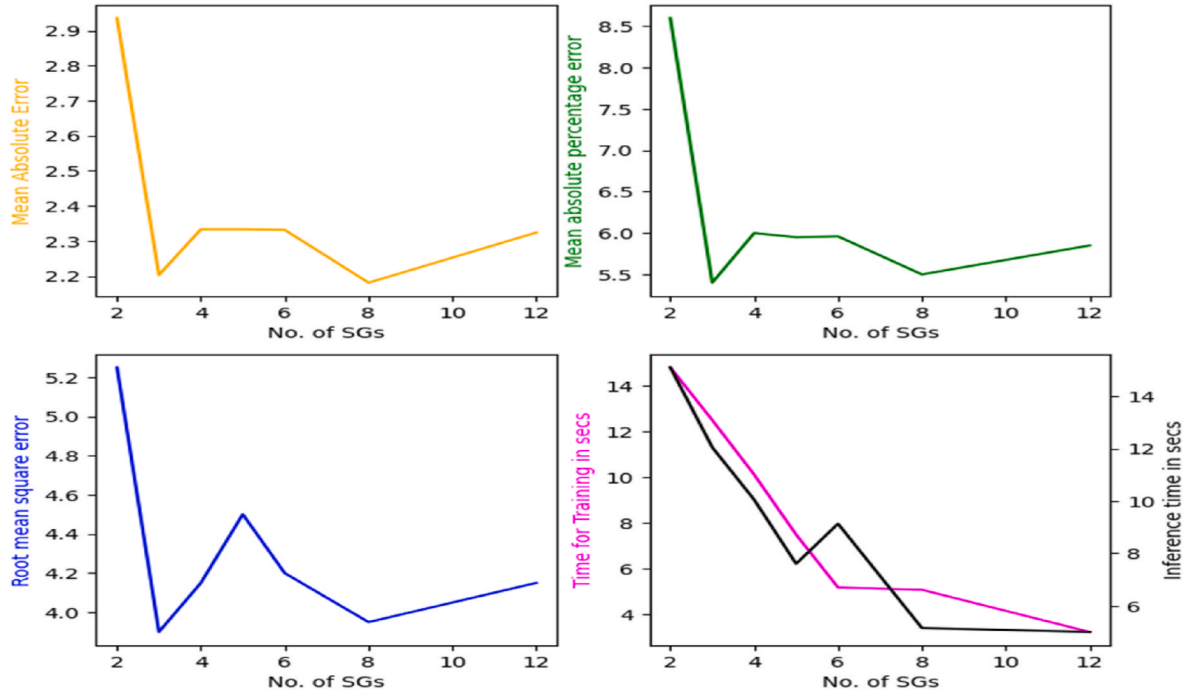


Fig. 4. Performance evaluation using METR-LA dataset for varying no. of connected spatial graphs in the validation set.

inference times. This observation suggests the possibility of achieving a harmonious balance between accuracy and computational efficiency by selecting an optimal value for this parameter. After careful deliberation, we determined that a value of eight for the number of spatial graphs in the METR-LA dataset offered the most favorable trade-off between accuracy and efficiency. Our analysis extended to the PEMS08 dataset, as depicted in Fig. 5, revealing variable performance of the FLAGCN model corresponding to the number of spatial graphs employed. In alignment with our approach for the METR-LA dataset, we employed eight spatial

graphs to construct the asynchronous spatial-temporal graph for the PEMS08 dataset. It's important to note that this process naturally introduced some degree of variability. These experiments were conducted meticulously, following a rigorous and technically sound methodology.

4.2. Number of sub graphs

The performance of the model is affected by the number of sub-

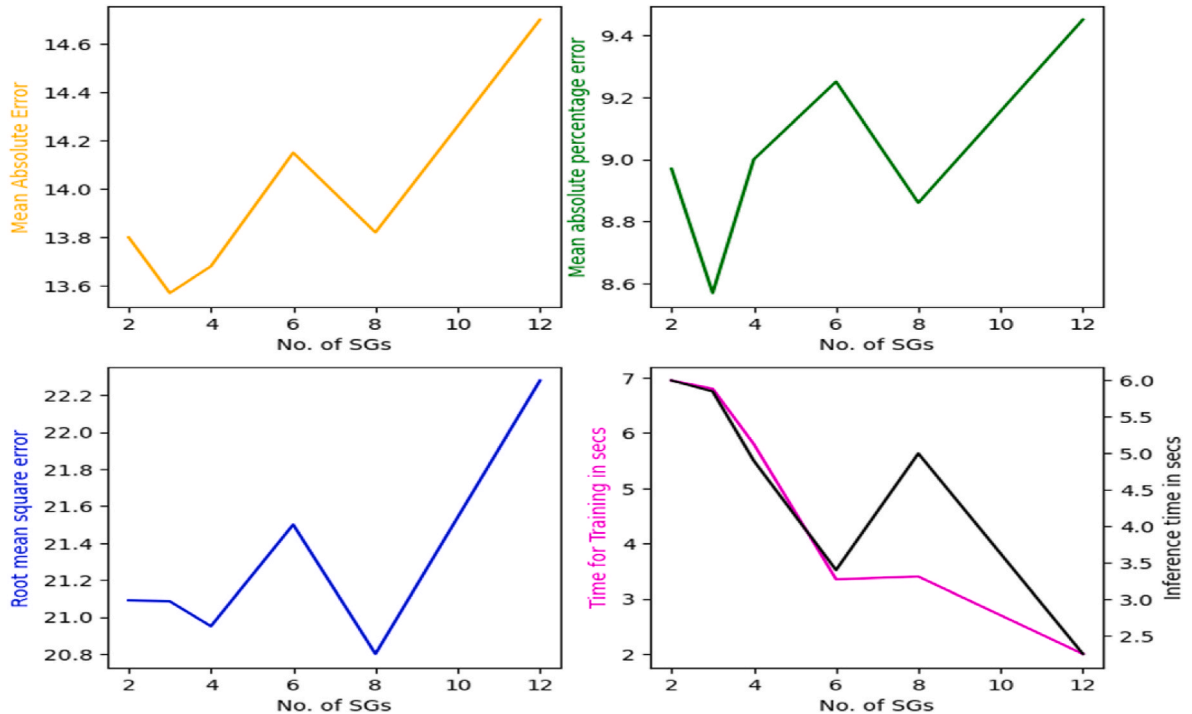


Fig. 5. Performance evaluation using PeMS08 dataset for varying numbers of connected spatial graphs within the validation set.

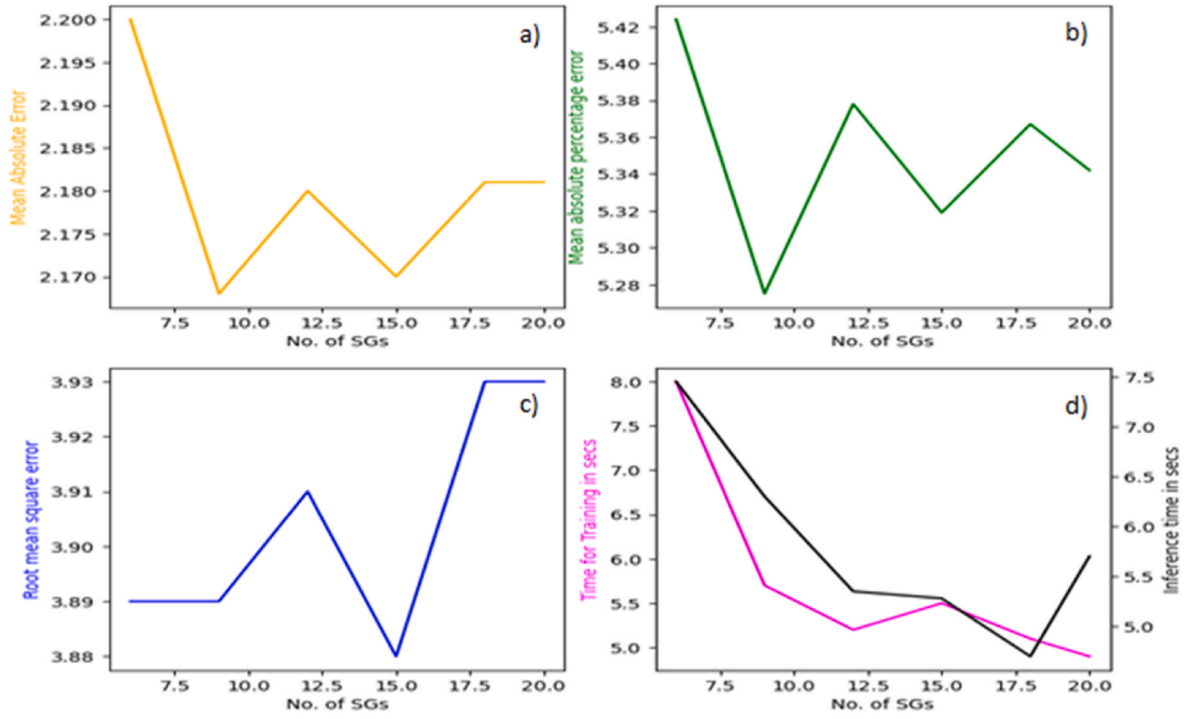


Fig. 6. Evaluation of performance on the METR-LA dataset across varying sub-graph counts.

graphs employed in our proposed FLAGCN strategy, which entails segmenting the traffic road network into numerous sub-graphs. Using the METR-LA dataset, we looked at six distinct sub-graph configurations: 6, 9, 12, 15, 18, and 20 sub-graphs, to evaluate various network partitioning strategies. Fig. 6(a), (b), and 6(c) illustrate our findings that the model's performance remained largely steady when the number of sub-graphs was equal to or more than 9. Additionally, as shown in Fig. 6(d), the computational time decreased when the number of sub-graphs was

lowered below 20. However, it is important to note that including each sub-graph requires a representative node with sufficient computational capacity, which affects the deployment costs of the model. Considering the trade-off between performance and cost, we have decided to partition the traffic road network into nine distinct sub-graphs specifically for the METR-LA dataset.

We investigated various network partitioning strategies, breaking the traffic road network into 4, 7, 10, 12, 13, 15, and 18 sub-graphs, in

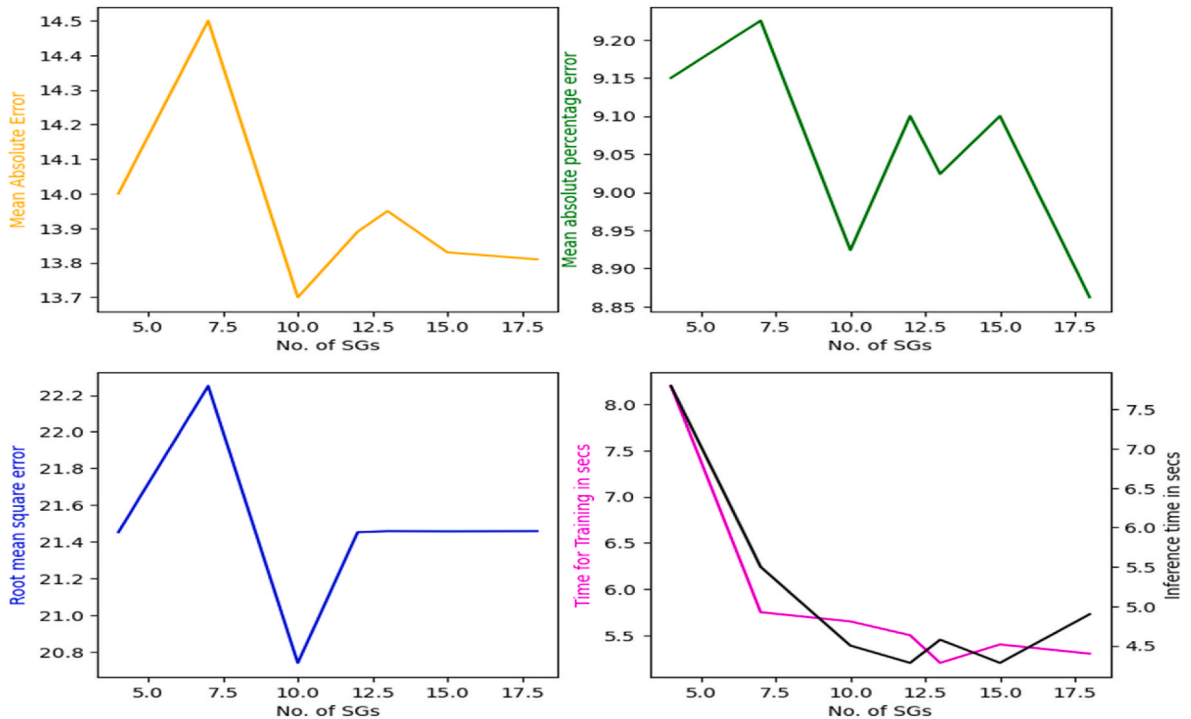


Fig. 7. Evaluation of performance on the PeMS08 dataset across varying sub-graph counts.

order to evaluate the performance of the PEMS08 dataset for traffic flow prediction as shown in Fig. 7. By examining the interplay between performance and cost considerations, we determined that utilizing 10 sub-graphs for the FLAGCN model offered optimal results. To ensure the FLAGCN model's performance was maximized, we employed a methodical grid search approach for parameter selection. Through this process, we identified the most suitable values for the learning rate (0.001), batch size (32), and training epochs (100). These parameters were carefully chosen to optimize the model's performance in traffic flow prediction experiments, striking a balance between accuracy and computational efficiency. By taking a thoughtful and strategic approach to partitioning the network and selecting model parameters, we were able to create a robust and effective FLAGCN model capable of accurately predicting traffic flow patterns in diverse scenarios.

Table 1 displays the outcomes of the forecasting analysis, including the time for training and inference, for both the FLAGCN method and the baseline methods in the context of one-step ahead predictions with a 5-min interval. The entries that are highlighted in bold indicate the models that have demonstrated superior performance. According to the findings presented in Table 1, the performance of the FLAGCN model in terms of prediction accuracy is on par with that of the most advanced models currently available in the field. In addition, it is worth noting that FLAGCN demonstrates a noteworthy decrease in training and inference durations compared to currently available deep-learning models. Within the context of the models presented in Table 1 it is observed that Graph Wavenet, T-GCN, GMAN and DCRNN, and are categorized as centralized deep learning models. Conversely, FLAGCN functions as a federated learning model. The models under consideration in this study incorporate the temporal correlation of traffic data and the spatial connections within the traffic road network.

In comparison to the T-GCN model, the GMAN model has enhanced prediction accuracy because it makes use of the attention mechanism to precisely capture the influence of temporal and spatial elements on traffic flow. Furthermore, the FLAGCN model exhibits prediction accuracy on par with cutting-edge deep learning models. The FLAGCN model outperforms the GMAN model in the setting of the METR-LA dataset, as shown by a lower mean absolute error (MAE) value of 2.17.

5. Limitations and future directions

5.1. Limitations

1. Validated only on highway traffic data (METR-LA/PEMS08); performance on urban networks with traffic signals/pedestrians remains untested.
2. Assumes fixed road topology, limiting adaptability to dynamic scenarios like construction zones or accidents.
3. Federated training assumes homogeneous client hardware; real-world edge devices vary in compute/energy capacity.

5.2. Future directions

1. Incorporate real-time graph rewiring to handle sudden topology changes (e.g., road closures).
2. Extend FLAGCN to jointly model vehicular flow, pedestrian data, and public transit for urban scenarios.
3. Develop adaptive aggregation strategies to accommodate heterogeneous edge devices.

6. Conclusion

Our proposed FLAGCN is an exceptional traffic flow prediction model that combines federated learning techniques with an enhanced Asynchronous Graph Convolutional Network (AGCN). It effectively addresses parameter update challenges in Graph Convolutional Networks (GCN), resulting in improved accuracy and reduced computational time

overhead. With tailored sub-graph partitioning for traffic networks and the novel Graph FL framework, FLAGCN outperforms existing centralized deep learning models on real traffic datasets. Its ability to continuously adapt, capture dynamic spatial-temporal relationships, and handle complex dependencies in the traffic network positions FLAGCN as a powerful tool for optimizing traffic management and planning. This model represents a significant advancement in traffic flow prediction and holds promising potential for mitigating traffic congestion and improving transportation infrastructure in the future.

CRedit authorship contribution statement

Muhammad Yaqub: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Shahzad Ahmad:** Writing – review & editing, Validation, Methodology. **Malik Abdul Manan:** Formal analysis, Data curation. **Muhammad Salman Pathan:** Writing – review & editing, Resources. **Lan He:** Writing – review & editing, Writing – original draft, Supervision.

Availability of data and materials

The datasets generated and/or analyzed during the current study are publicly available on the following link, <https://drive.google.com/drive/folders/10FOTa6HXPqX8Pf5WRoRwcFnW9BrNZEIX>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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